

Max Ortner

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WORK EXPERIENCE

Los Alamos National Laboratory

June, 2023 - Present

Post-Baccalaureate Researcher

Los Alamos, NM

- Worked in a large development team on the C++, open-source software FleCSI
 - Reduced MPI memory overhead on an algorithm within FleCSI(*publication pending*) by 50% by leveraging HPC systems and profilers to identify inefficiencies
 - Ran protocols and maintained a development environment over SSH in a command-line Linux environment; used large scale parallel schedulers like Slurm and dependency managers like Spack
 - Managed issue tracking, participated in code review, and oversaw software releases through Gitlab
- Collaborated with scientists in the theoretical and material science divisions to make their physics software performant, at one time reducing a large scale, python machine learning algorithm's wall time by 120x
- Proven experience using CLI version control software within a collaborative team setting on sensitive, large-scale, ITAR compliant software

PROJECTS

Quadcopter Digital Twin

June, 2024 - Present

Personal

maxortner.com/uav

- Created a simple scheme to test various quadcopter flight software implementations by running a full 6-DOF physics simulation and storing the results in an HDF5 file
- Combined several personal libraries to create tools used to visualize 3D telemetry data in order to diagnose issues and tune control system parameters

Combustion Hydrodynamics Solver

July, 2024 - Present

Los Alamos National Laboratory

- Created an adaptive spatial discretization method for the mesh to efficiently solve hydro PDEs on various resolution scales
- Used CUDA and cuSOLVER to maximize throughput, which allowed for extreme performance gains on the lab's large computer systems as well as small-scale throughput optimized devices

Simple Lua

January, 2024 - May, 2024

Personal

simple-lua.maxortner.com

- C++ wrapper for the Lua scripting language that prioritizes ease-of-use and utilizes the robust C++ type system to communicate with the Lua runtime
- Thorough CI and documentation to support easy integration and code development

Midnight Graphics

May, 2024 - Present

Personal

mn.maxortner.com

- Cross-platform 3D graphics library with a simple interface
- Utilizes the Vulkan graphics API to maximize performance potential

EDUCATION

Belmont University

December, 2023

Bachelor of Science, Math and Physics

Nashville, TN

- Graduated *magna cum laude* with a 3.75 GPA
- Completed 5 semesters of research in physics and math (one of which won an institutional award)
- Started the physics club and was voted in as president